

RAJESH NAUTIYAL

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Senior Professional II - Infrastructure Validation, Automation & AI-Assisted Engineering

PROFESSIONAL SUMMARY

Senior Technical Lead with 14+ years across Cisco enterprise networking, infrastructure validation, and regression engineering. Led validation environments of 150+ devices and 500+ Python/PyATS automation scripts with Jenkins-based CI execution across NXOS and IOS-XE platforms. Since Jan 2025, focused on AI-assisted engineering using Codex, GitHub Copilot, domain-specific RAG (MdRAG), structured bot instructions, and conversational assistants to improve RCA, topology validation, infrastructure readiness, code remediation, and engineering productivity.

CORE SKILLS

AI-Assisted Engineering: Codex, GitHub Copilot, Prompt Engineering, Agent-Assisted Workflow Design, Domain-Specific RAG, AI-Based RCA, Conversational Knowledge Systems, AI-Driven Troubleshooting

Automation and Development: Python, PyATS, Shell Scripting, YAML, JSON, TCL

Network Platforms and Protocols: Cisco NXOS, IOS-XE, VXLAN, BGP, OSPF, EIGRP

Programmability and APIs: YANG Models, REST APIs, API Response Validation, JSON/YAML Parsing

Infrastructure and Tools: Linux RHEL7/RHEL8, Cisco UCS, VMware ESXi, AWS, Virtualized Network Labs, Jenkins, Git, CVS, Visual Studio Code

PROFESSIONAL EXPERIENCE

CAPGEMINI | Dec 2022 - Present

Senior Professional II | Infrastructure Validation, Automation & AI-Assisted Engineering | Project: Cisco

AI-Assisted Engineering and Automation Initiatives | Jan 2025 - Present

- Reduced failure-triage and RCA investigation time from 1-2 hours to ~15 minutes by building AI-assisted workflows on an internal domain-specific RAG (MdRAG).
- Designed and deployed an AI-powered regression support assistant giving engineers on-demand access to regression insights, troubleshooting guidance, and execution status.
- Created reusable AI skillsets and structured bot instructions enabling engineers to access validation workflows and operational knowledge through conversational interfaces.
- Built AI-assisted topology validation and infrastructure-readiness solutions covering interface/CDP verification, connectivity checks, device cleanup, and baseline provisioning.
- Accelerated script development, repository analysis, code remediation, and debugging using OpenAI Codex, GitHub Copilot, and AI-assisted development workflows.
- Developed a Django-based Cisco UCS infrastructure-management application (Codex-assisted and validated) supporting virtual device lifecycle management and operational monitoring.

Infrastructure Validation and Regression Leadership | Dec 2022 - Present

- Lead regression and validation for Cisco NXOS and IOS-XE platforms, collaborating directly with Cisco engineering teams on enterprise networking releases.
- Lead a team of 5 engineers across regression execution, script maintenance, failure triage, release validation, defect follow-up, and quality sign-off.
- Manage validation infrastructure of 150+ physical and virtual devices and 500+ automation scripts supporting multiple parallel release cycles.
- Own lab readiness, topology creation, hardware allocation, Linux administration, Python environments, and end-to-end Jenkins-based validation workflows.

TATA CONSULTANCY SERVICES (TCS) | Mar 2012 - Dec 2022

Technical Lead | Mar 2018 - Dec 2022 | Project: Cisco

- Led a team of 6 engineers across regression triage, automation execution, defect analysis, and release validation for Cisco NXOS and IOS-XE platforms.
- Designed and managed CI/CD-based regression workflows, including Jenkins job execution, regression scheduling, failure tracking, and release-readiness reporting.
- Owned complex regression debugging and escalation handling by analyzing automation scripts, framework behavior, parser outputs, device logs, and protocol states.
- Partnered with PyATS development teams to validate new and enhanced automation libraries; identified framework gaps and parser issues, raising enhancement requests with reproduction steps and recommended fixes.
- Developed and maintained Python/PyATS validation scripts for protocol testing, configuration validation, telemetry validation, API response verification, and regression workflows.
- Designed automation flows for model-driven programmability validation using YANG models, REST APIs, and JSON/YAML parsing.
- Mentored team members on PyATS workflows, CI/CD regression execution, script debugging, and Cisco platform troubleshooting.

Automation Engineer | May 2012 - Feb 2018 | Project: Cisco

- Developed and maintained TCL-based validation scripts and regression parsers for router/switch command verification across Cisco NXOS and IOS-XE platforms.
- Enhanced TCL logic for command execution, output parsing, result comparison, error handling, and regression reporting.
- Debugged TCL scripts, parser failures, and execution errors during Cisco platform release cycles.
- Analyzed device logs, parser outputs, and protocol states to identify root causes and support defect triage.

SELECTED PROJECTS

AI-Powered Chat Bot - Failure Triage and RCA Solution: Built an AI-assisted solution analyzing regression failures, traceback errors, device logs, and historical defect data to suggest probable root causes, debugging steps, and known-issue correlations; significantly reduced manual triage effort.

Tech Stack: Webex Chat, Open AI Codex, internal MdRAG Integration, SCRBOT API Configuration, Red Hat Server

AI-Driven UCS Operations Assistant: Designed AI workflows using domain-specific instructions and internal knowledge systems to support regression insights, topology validation, infrastructure operations, and troubleshooting recommendations.

Tech Stack: Open AI Codex, Python, Internal MdRAG

KEY ACHIEVEMENTS

- Reduced regression failure-triage time from 1-2 hours to ~15 minutes using AI-assisted analysis.
- Managed infrastructure of 150+ devices and 500+ regression scripts across Cisco enterprise networking platforms.
- Completed 300+ automation integration activities across Cisco platform validation programs.
- Designed and deployed AI-powered regression support and failure-analysis solutions adopted by engineering teams for troubleshooting and operational decisions.
- Trained and mentored 10+ engineers on validation workflows, AI-assisted troubleshooting, and PyATS/TCL automation.
- Recognized with Certificate of Excellence (TCL to PyATS Migration), Special Initiative Award, and Star of the Month Award.

EDUCATION AND CERTIFICATION

Education: Bachelor of Technology (B.Tech) in Information Technology - GB Pant University of Agriculture and Technology, 2011

Certification: AWS Certified Developer - Associate, 2022